

PROGRAMS

```
public class Factorial {  
  
    public static void main(String[] args) {  
  
        int num = 10;  
        long factorial = 1;  
        for(int i = 1; i <= num; ++i)  
        {  
            // factorial = factorial * i;  
            factorial *= i;  
        }  
        System.out.printf("Factorial of %d = %d", num, factorial);  
    }  
}
```

Output

```
Factorial of 10 = 3628800  
=== Code Execution Successful ===
```

```
2 . import java.math.BigInteger;
```

```
public class Factorial {  
  
    public static void main(String[] args) {  
  
        int num = 30;  
        BigInteger factorial = BigInteger.ONE;  
        for(int i = 1; i <= num; ++i)  
        {  
            // factorial = factorial * i;  
            factorial = factorial.multiply(BigInteger.valueOf(i));  
        }  
        System.out.printf("Factorial of %d = %d", num, factorial);  
    }  
}
```

```
}
```

```
Factorial of 30 = 2652528598121910586363084800000000  
=== Code Execution Successful ===
```

```
3. public class MultiplicationTable {  
  
    public static void main(String[] args) {  
  
        int num = 5;  
        for(int i = 1; i <= 10; ++i)  
        {  
            System.out.printf("%d * %d = %d \n", num, i, num * i);  
        }  
    }  
}
```

```
Output  
5 * 1 = 5  
5 * 2 = 10  
5 * 3 = 15  
5 * 4 = 20  
5 * 5 = 25  
5 * 6 = 30  
5 * 7 = 35  
5 * 8 = 40  
5 * 9 = 45  
5 * 10 = 50  
  
=== Code Execution Successful ===
```

```
4. public class MultiplicationTable {  
  
    public static void main(String[] args) {  
  
        int num = 9, i = 1;  
        while(i <= 10)  
        {  
            System.out.printf("%d * %d = %d \n", num, i, num * i);  
            i++;  
        }  
    }  
}
```

```

    }
}
}

```

Output

```

9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90

```

=== Code Execution Successful ===

```

5. class Main {
    public static void main(String[] args) {

        int n = 10, firstTerm = 0, secondTerm = 1;
        System.out.println("Fibonacci Series till " + n + " terms:");

        for (int i = 1; i <= n; ++i) {
            System.out.print(firstTerm + ", ");

            // compute the next term
            int nextTerm = firstTerm + secondTerm;
            firstTerm = secondTerm;
            secondTerm = nextTerm;
        }
    }
}

```

Output

```

Fibonacci Series till 10 terms:
0, 1, 1, 2, 3, 5, 8, 13, 21, 34,
=== Code Execution Successful ===

```

```

6.class Main {

```

```

public static void main(String[] args) {

    // find GCD between n1 and n2
    int n1 = 81, n2 = 153;

    // initially set to gcd
    int gcd = 1;

    for (int i = 1; i <= n1 && i <= n2; ++i) {

        // check if i perfectly divides both n1 and n2
        if (n1 % i == 0 && n2 % i == 0)
            gcd = i;
    }

    System.out.println("GCD of " + n1 + " and " + n2 + " is " + gcd);
}
}

```

Output

GCD of 81 and 153 is 9

=== Code Execution Successful ===

```

7.class Main {
    public static void main(String[] args) {

        // find GCD between n1 and n2
        int n1 = 81, n2 = 153;

        while(n1 != n2) {

            if(n1 > n2) {
                n1 -= n2;
            }

            else {
                n2 -= n1;
            }
        }
    }
}

```

```
    System.out.println("GCD: " + n1);  
  }  
}
```

GCD: 9

=== Code Execution Successful ===

```
8. public class Main {  
  
    public static void main(String[] args) {  
  
        int count = 0, num = 0003452;  
  
        while (num != 0) {  
            // num = num/10  
            num /= 10;  
            ++count;  
        }  
  
        System.out.println("Number of digits: " + count);  
    }  
}
```

Number of digits: 4

=== Code Execution Successful ===

```
9. public class Main {  
  
    public static void main(String[] args) {  
  
        int count = 0, num = 123456;
```

```
for (; num != 0; num /= 10, ++count) {  
    }  
  
    System.out.println("Number of digits: " + count);  
}  
}
```

Number of digits: 6

=== Code Execution Successful ===

```
10. public class Main {  
  
    public static void main(String[] args) {  
  
        // year to be checked  
        int year = 1900;  
        boolean leap = false;  
  
        // if the year is divided by 4  
        if (year % 4 == 0) {  
  
            // if the year is century  
            if (year % 100 == 0) {  
  
                // if year is divided by 400  
                // then it is a leap year  
                if (year % 400 == 0)  
                    leap = true;  
                else  
                    leap = false;  
            }  
  
            // if the year is not century  
            else  
                leap = true;  
        }  
  
        else  
            leap = false;  
    }  
}
```

```
if (leap)
    System.out.println(year + " is a leap year.");
else
    System.out.println(year + " is not a leap year.");
}
```

```
1900 is not a leap year.
```

```
=== Code Execution Successful ===
```